BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

One-Dimensional Arrays

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does input validation check?

2. Valid ages are integers from 13 to 18. Which input has the right type but is invalid?

A) 12 B) 13
C) 15 D) 18

3. "Defensive design assumes every user always enters valid and harmless data." Correct this. Explain why.

4. A desktop needs a LAN connection. "A NIC keeps files when power is off." Correct this. What does a NIC do here?

Today's objectives

Create arrays and access their elements. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

1D array
A fixed-length indexed collection holds values of one data type.

Index
Array positions start at zero, so index 2 refers to the third element.

Element
An element is a stored item accessed by its array index.

Assignment
An array element can be updated by assigning to its position.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

1D array	A fixed-length indexed collection holds values of one data type. Apply this in a five sports scores.
Index	Array positions start at zero, so index 2 refers to the third element. Apply this in a library loans.
Element	An element is a stored item accessed by its array index. Apply this in a weekly temperatures.
Assignment	An array element can be updated by assigning to its position. Apply this in a stock levels.

I Do - One worked example

A five sports scores is being developed. Explain how 1d array and index are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. 1D array [1]; A fixed-length indexed collection holds values of one data type. [1].

3. Index [1]; Array positions start at zero, so index 2 refers to the third element. [1].

4. Two linked points for each concept.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A library loans uses 1d array. Explain how it helps. [2 marks]

2. Explain index in a weekly temperatures. [2 marks]

3. A stock levels uses element and assignment. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write one line to print the last value in this zero-indexed array.

```
array driveGB = [64,128,256]
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how index and element help with a quiz marks. [4 marks]

Q4. A programmer designs a club attendance. Explain why 1d array, element and assignment matter. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An array is an indexed collection of elements of the same data type. ☐ T ☐ F
2. A fixed-length indexed collection holds values of one data type ☐ T ☐ F
3. Array positions start at zero, so index 2 refers to the third element ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Whether input meets specified rules; it does not prove factual truth or identity. ☐

Q2 A - 12 ☐

Q3 Planning for invalid, unexpected or malicious use and handling it safely. ☐

Q4 A NIC lets a device connect to a network. It has a different job from storage or a router. ☐

Code First - Model

```
array driveGB = [64,128,256]
print(driveGB[2])
```

Output: 256

We Do - Check the model

1. 1D array [1]; A fixed-length indexed collection holds values of one data type. [1]. ☐

2. Index [1]; Array positions start at zero, so index 2 refers to the third element. [1]. ☐

3. Element [1]; An element is a stored item accessed by its array index. [1]. Assignment [1]; An array element can be updated by assigning to its position. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 1D array [1]; Element [1]. ☐

Q3 Index [1]; Array positions start at zero, so index 2 refers to the third element. [1]. Element [1]; An element is a stored item accessed by its array index. [1]. ☐

Q4 1D array [1]; A fixed-length indexed collection holds values of one data type. [1]. Element [1]; An element is a stored item accessed by its array index. [1]. Assignment [1]; An array element can be updated by assigning to its position. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: